

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-402 — Probability and Statistics

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt any two questions from Section A, any six questions from Section B, and all questions from Section C. The marks for each question are given against the respective questions.

SECTION — A**Question 1.**

- (a) State and prove Markov's inequality. [3]
- (b) Derive Chebyshev's inequality using Markov's inequality. [3]
- (c) Let $f(x)$ be the uniform distribution on $0 \leq x \leq 5$ and 0 everywhere else. Give a bound using Chebyshev's inequality for $P(1 \leq X \leq 4)$ and also calculate the actual probability. How do they compare? [6]

Question 2.

- (a) Give the axiomatic definition of probability. [2]
- (b) Define convergence in distribution. [2]
- (c) Consider the joint probability mass function of two discrete random variables X and Y as $f(x, y) = \frac{xy^2}{13}$ with support $\{(x, y) = (1, 1), (1, 2), (2, 2)\}$. Construct a joint probability distribution table of X and Y . Are X and Y independent? Find $E(X)$ and $E(Y)$. [8]

Question 3.

- (a) Define simple and composite hypothesis with examples. [2]
- (b) What are Type-I and Type-II errors in hypothesis testing? [2]
- (c) Urban storm water can be contaminated by many sources, including discarded batteries. When ruptured, these batteries release metals of environmental significance. A sample of 49 Panasonic AAA batteries gave a sample mean zinc mass of 2.06 g and a sample standard deviation of 0.140 g. Does this data provide compelling evidence for concluding that the population mean zinc mass exceeds 2.0 g? Construct a 95% confidence interval for the population mean zinc mass. Choose the appropriate critical value(s) ($z_{0.05} = 1.64, t_{0.05, 48} = 1.68, t_{0.025, 48} = 2.01, z_{0.025} = 1.96$). [8]

Question 4. Let X be a random variable with PDF given by:

$$f_X(x) = \begin{cases} x^2(2x + c) & \text{if } 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of c . Also compute the mean, variance and CDF of the random variable X . If $Y = \frac{2}{x} + 3$ is a function of the random variable X , then find $E(Y)$ and $\text{Var}(Y)$. [12]

SECTION — B

Question 5.

- (a) If the random variables X and Y independently follow exponential distribution with parameters θ and λ , respectively, then find the expectation and variance of $Z = X + Y$. [4]
- (b) What do you mean by standardization of a variable? [2]

Question 6.

- (a) Show that the Pearson's correlation coefficient between the variables X and Y is independent of change of scale and origin. [4]
- (b) Assume neither A nor B is an impossible event. If A and B are mutually exclusive, will they be independent? [2]

Question 7.

- (a) Show that the Poisson distribution is a limiting case of the Binomial distribution. [4]
- (b) What is the p-value in hypothesis testing? [2]

Question 8. A person uses his car 30% of the time, walks 30% of the time and rides the bus 40% of the time as he goes to work. He is late 10% of the time when he walks; he is late 3% of the time when he drives; and he is late 7% of the time when he takes the bus. What is the probability he took the bus if he was late? What is the probability he walked if he is on time? What is the probability that he'll be on time? [6]

Question 9. Vehicles pass through a junction on a busy road at an average rate of 240 per hour. Find the probability that none passes in a given minute. What is the expected number passing in three minutes? Find the probability that this expected number actually passes through in a given three-minute period. (Assume $e \approx 2.72$). [6]

Question 10.

- (a) State and prove the law of total probability. [4]
- (b) Give an example where two events are mutually exclusive but not exhaustive. Also give an example where two events are not mutually exclusive but exhaustive. [2]

Question 11. Define Normal distribution. Also discuss its important properties. [6]

Question 12.

- (a) Suppose the average CGPA of students is normally distributed with population mean 8.7 and standard deviation 0.2. At 10% level of significance, what is the minimum sample size needed to obtain a margin of error of 0.013? [4]

- (b) What is a cumulative distribution function? What are the criteria that a valid CDF needs to satisfy? [2]

Question 13. The results of a national survey showed that on average, adults sleep 6.9 hours per night. Suppose that the standard deviation is 1.2 hours. Use Chebyshev's theorem to calculate the minimum percentage of individuals who sleep between 4.5 and 9.3 hours. Use Chebyshev's theorem to calculate the minimum percentage of individuals who sleep between 3.9 and 9.9 hours. Assume that the number of hours of sleep follows a normal distribution. Use the empirical rule to calculate the percentage of individuals who sleep between 4.5 and 9.3 hours per day. How does this result compare to the value that you obtained using Chebyshev's theorem in the first part of this question? [6]

SECTION — C

Question 14. Answer all the questions in a sentence each: [10x1=10]

- (a) What is the sampling distribution of sample variance?
- (b) What is the range of Pearson's correlation coefficient?
- (c) When will the central limit theorem fail, even if the sample size is large?
- (d) What is the relation between power of the test and Type-II error?
- (e) Construct a sample space for the event of tossing a coin till the first head appears.
- (f) Let $P(B) > 0$ so that $P(A|B)$ is defined. If $A \subseteq B$, what is $P(A|B)$?
- (g) Define probability mass function with a suitable example.
- (h) How can we get the binomial distribution from Bernoulli random variables?
- (i) If mean = median = mode, what can we say about that distribution?
- (j) What is the formula for sample covariance?